Vision-Based Precision Pose Estimation for Autonomous Formation Flying

UE24CS352A Machine Learning Mini-Project | Section J, Group 16, Problem #9 | Neha K S (PES1UG24CS581), Mohith N (PES1UG24CS580)

1. Problem statement

A follower aircraft F carries a camera that is gimballed to always point at a leader aircraft L. From a single image we must estimate the relative pose of the leader (position and roll about the line of sight) precisely enough for *tight* formation flight. Our assigned problem is the reproduction of Punnoose (Stanford CS229, 2020): an ANN classifies detected aircraft features into a coarse discrete pose label, and a particle filter (PF) informed by that label refines it. GPS/inertial sensors are not precise enough at these separations, and a pure end-to-end regressor gives no uncertainty and ignores dynamics, which motivates the hybrid design.

2. Dataset

The paper releases no data, so we generate it synthetically. A 14-feature aircraft model (nose, canopy, wing/stabiliser tips, nozzles, intakes, ...) is projected through a pinhole camera (1280x960 px, f = 1600 px). Each feature has an outward normal; features facing away from the camera are *occluded* (a normal-based stand-in for ray tracing). The detector adds 0.5 px Gaussian noise and misses 2 % of visible features. A sample is the 42-vector [u, v, occluded] x 14, and its label is one of **4800** pose cells (x: 10 bins over [-100, 50] m, y: 10 over [-50, 50] m, z: 8 over [-20, 20] m, roll: 6 over [-45, 45] deg, i.e. Table I of the paper). We draw 300 000 training, 30 000 validation and 30 000 test poses uniformly in that box (range $\geq$ 15 m). Evaluation uses 10 random 10 s approach trajectories (10 Hz) that the follower flies from ~90 m to ~30 m behind the leader while the leader manoeuvres.

3. Approach

(a) ANN pose-label classifier: 42 $\rightarrow$ 100 $\rightarrow$ 100 $\rightarrow$ 4800, ReLU, softmax, Adam (as in the paper). **(b) Particle filter**: particles are initialised from the ANN label plus a (true – label) error pool measured on validation data. Each step, particles are propagated through the relative dynamics, where the leader's unknown speed, pitch and roll appear as a uniform disturbance; each particle's 14 features are projected and compared with the detected ones (occluded ones ignored); particles are weighted, and the resampling draws floor(αN) from the weighted set and the rest from the ANN label + error pool. We implement the paper's inverse-MSE likelihood (two configurations: N=1000, α=0.9 every 10 cycles; N=5000, α=0.9 every cycle) and a Gaussian likelihood with roughening (N=2000, α=0.5). **(c) Our extension** (the paper's own future-work suggestion): a Levenberg-Marquardt minimisation of the reprojection error over the 4-DoF pose, started from the ANN's top-3 classes and keeping the lowest-cost solution (single-view bundle adjustment).

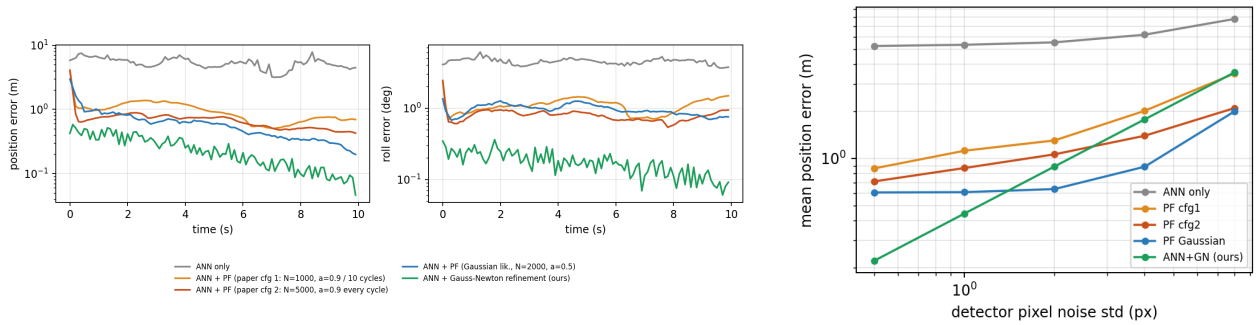
4. Implementation overview

Python + numpy/scipy/matplotlib only (no DL framework: forward/backward passes of the ANN are written by hand). Package `poseest/`: `aircraft.py` (model, projection, occlusion, detector), `discretize.py`, `dataset.py`, `ann.py`, `dynamics.py`, `particle_filter.py`, `refine.py`, `pipeline.py`. Scripts: `train_ann.py` (~9 min on one CPU core), `run_experiment.py`, `noise_sweep.py`, `demo.py` (live demo + GIF). Unit tests cover label round-trips, projection scaling, occlusion, the refinement and trajectory validity.

5. Results

Method	Position error (m)	Roll error (deg)
ANN only (cell centre)	5.32 ± 0.31	4.51 ± 0.58
ANN + PF, paper cfg 1 (N=1000)	0.96 ± 0.25	1.09 ± 0.20
ANN + PF, paper cfg 2 (N=5000)	0.70 ± 0.09	0.80 ± 0.14
ANN + PF, Gaussian likelihood (N=2000)	0.62 ± 0.09	0.96 ± 0.09
ANN + Gauss-Newton refinement (ours)	**0.25 ± 0.04**	**0.18 ± 0.03**

Mean ± std over 10 trajectories. ANN test accuracy: top-1 69.6 % (paper: 54 %), top-5 96.0 %, 98.3 % within one neighbouring cell; mean cell-centre error 5.6 m / 4.3°, so almost all error is discretisation, as the paper observed.



Left: mean error over time (log scale). Right: robustness to detector pixel noise (ANN not retrained).

6. Conclusions

The ANN gives a reliable *coarse* label (70 % exact, 98 % within one cell) but its precision is capped by the 15 m x 10 m x 5 m x 15° cell size. Unlike the paper, where the PF changed little, our ANN-informed PF cuts the error about 7x (5.3 m $\rightarrow$ 0.6-0.7 m); we attribute this to our cleaner simulator (exact camera model, Gaussian noise, no outliers, 4-DoF state), so absolute numbers are **not** comparable to the paper. The reprojection-based refinement is clearly best at low noise (0.25 m, 0.18°), confirming the paper's suggestion that the ANN is best used as a robust initialiser. It is per-frame, though: beyond ~2 px of detector noise the

temporal Gaussian PF overtakes it, so the natural next step is to fuse both (refinement as the PF measurement). Limitations: synthetic data only, simplified occlusion and dynamics, and the CNN feature detector is assumed rather than built.